CIVICARE – AI POWERED PLATFORM FOR CITIZEN COMPLAINTS & INFRASTRUCTURE MAINTENANCE

A THESIS

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Ms. Trupti Kantale

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PROCEEDINGS OF EXTERNAL VIVA-VOCE EXAMINATION

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ABSTRACT

Rapid urbanization has significantly burdened municipal corporations, leading to inefficient civic grievance redressal mechanisms. Traditional systems suffer from manual processing delays, a lack of transparency, and a high volume of spurious or non-contextual complaints (e.g., selfies, downloaded images). This project presents CiviCare, an innovative, intelligent end-to-end web portal designed to revolutionize smart city governance.

The proposed system integrates advanced generative artificial intelligence (Google Gemini 1.5 Flash) as a strict anti-spoofing gatekeeper, combined with a natively built Heuristic Natural Language Processing (NLP) module for deterministic auto-routing. To ensure data integrity, the system enforces live camera captures and applies cryptographic visual watermarking (embedding real-time GPS coordinates and timestamps directly onto the image pixels). Furthermore, the system employs recursive geofencing via the OpenStreetMap Overpass API to automatically prioritize issues located near critical infrastructure, such as hospitals and schools.

The backend architecture, built on ASP.NET (C#) and Microsoft SQL Server, enforces Role-Based Access Control (RBAC) with a separated secure gateway for municipal staff (Water, Electricity, and Sanitation departments) and a public portal for citizens. CiviCare also features a dual-asynchronous Simple Mail Transfer Protocol (SMTP) email engine that broadcasts live status updates to citizens and task assignments with Google Maps links to field workers. By automating the verification, routing, and communication pipelines, CiviCare drastically eliminates fake grievance loads and fosters transparent, responsive urban governance.

Keywords: E-Governance, Natural Language Processing (NLP), Global Positioning System (GPS), Civic Grievance System, Automated Rou

CHAPTER 1

INTRODUCTION

The paradigm of "Smart Cities" relies heavily on smart governance and active citizen participation. However, one of the primary hurdles in achieving operational excellence in urban centers is the failure of existing civic grievance redressal systems. Currently, municipalities rely on traditional web portals or manual registers that require heavy human intervention for sorting, validating, and routing complaints. Consequently, citizens face significant delays and a pervasive lack of transparency regarding the status of their grievances.

Furthermore, a critical challenge faced by Urban Local Bodies (ULBs) is verifying the authenticity of reported issues. Without strict validation, databases are often flooded with spam, selfies, or downloaded images, wasting valuable administrative bandwidth.

Addressing these critical gaps, this project introduces CiviCare, a centralized, AI-assisted grievance management portal. CiviCare replaces manual filtering with a "Hybrid Validation Architecture." It utilizes Google Gemini AI strictly as a gatekeeper to block fraudulent uploads and employs a highly accurate Keyword NLP engine to route legitimate issues based on localized semantics.

By separating the public citizen interface (featuring Google OAuth 2.0) from the highly secure, restricted staff gateway, the system ensures maximum security. Through automated live tracking, dynamic dashboards, and public analytics, CiviCare establishes a transparent, accountable, and high-efficiency standard for modern municipal maintenance.

1.1 Problem Statement

In most cities today, the way citizens report daily problems like broken streetlights, leaking water pipes, or massive garbage piles to the municipal corporation is slow, confusing, and highly inefficient. The current manual systems create a massive gap between the public and the government due to several major issues:

Fake and Spam Complaints Waste Time: Many people upload fake photos, selfies, downloaded internet pictures, or old photos from their galleries. Because city workers have to manually check every single complaint, they end up wasting hours visiting locations where no actual problem exists. There is no system to verify if a photo is genuine and recently taken at the exact location.

Manual Sorting and Wrong Departments: When a citizen reports an issue, they often don't know exactly which department (Water, Electricity, or Sanitation) is supposed to fix it. Manual sorting by government clerks takes days, and complaints frequently get forwarded to the wrong department, causing huge delays in getting the problem fixed.

No Emergency Prioritization: Currently, all complaints are treated equally in a long queue. If a dangerous live wire falls near a school or a hospital, the city workers might not realize it is an emergency because they don't have an automated way to map the complaint's location against critical public areas.

Lack of Transparency and Public Frustration: Once a citizen submits a complaint, it goes into a "black hole." They get no live updates, no tracking IDs, and no idea if a field worker has been assigned to their area. Similarly, when the city plans a power cut or water shut down for maintenance, citizens are rarely informed in advance, leading to public panic and frustration.

1.2 Objectives Of the Study

1. Stop Fake Complaints: Automatically block invalid photos (like selfies or indoor pictures) using AI so that city workers do not waste time on fake issues.

2. Auto-Assign Departments: Read the citizen's complaint description and automatically send it to the correct department (Water, Electricity, or Sanitation) based on the words used.

3. Ensure Real Proof: Force users to take live camera photos and automatically stamp the exact GPS location, date, and time directly on the image so it cannot be faked.

4. Prioritize Urgent Issues: Automatically detect if a reported problem is near important places like schools or hospitals and instantly mark it as high priority.

5. Keep Citizens Updated: Send automatic email alerts to citizens at every step of their complaint and provide field workers with direct Google Maps links to fix the issue faster.

6. Send Maintenance Alerts: Allow city departments to schedule upcoming repair work (like planned power cuts or water shutoffs) and instantly email citizens so they are informed in advance.

1.3 Scope of the project

The scope of the CiviCare project defines what the system is built to do, who it is built for, and the boundaries within which it operates. The project is designed to be a complete, web-based digital bridge between the citizens of a city and the municipal corporation.

The scope of this project is divided into the following key areas:

1. Target Audience

The system is scoped to serve three main types of users:

Citizens: To easily report civic issues, track their live status, and receive advance alerts about planned maintenance (like power cuts or water supply stops).

Department Staff & Field Workers: Specifically limited to three core municipal departments—Electricity, Water Supply, and Sanitation. Staff can view assigned complaints, dispatch field workers, and mark issues as resolved.

Super Admin: To oversee the entire city's operations, manage user accounts, and onboard new department heads.

2. Functional Scope

The primary functions included within the boundaries of this project are:

Strict Issue Reporting: Allowing citizens to report problems using their mobile cameras. The system strictly captures live GPS coordinates and permanently prints a timestamp and location watermark on the photo to prevent fake uploads.

a. AI-Powered Gatekeeping: Using Google Gemini AI strictly to scan uploaded photos and instantly reject invalid images (like selfies, indoor photos, or blank screens) before they reach the database.
b. Smart Auto-Routing (NLP): Automatically reading the citizen's text description and using keyword matching (e.g., "leak", "wire", "garbage") to send the complaint directly to the correct department without human sorting.

c. Automated Emergency Prioritization: Using map data (Geofencing) to automatically detect if a reported problem is near a school or hospital, and immediately marking it as "High Priority".

Dual Email Communication: Sending automated email alerts to citizens (for OTP login, complaint tracking, and maintenance alerts) and to field workers (for task assignments with Google Maps location links).

Public Analytics: Providing a transparent dashboard where any citizen can see the total number of reported, active, and resolved complaints in the city.

3. System Boundaries

To keep the project focused and efficient, certain features are explicitly outside the scope of this system:

a. No Native Mobile App: The system is a mobile-responsive web portal, not a downloadable Android or iOS app. Therefore, it requires an active internet connection and does not work offline.

b. Limited Departments: The routing and management are strictly limited to Electricity, Water, and Sanitation. It does not handle police, medical emergencies, or traffic complaints.

c. Email-Only Notifications: All system alerts, OTPs, and tracking updates are sent exclusively via Email. The project does not include SMS, WhatsApp, or mobile push notifications.

CHAPTER 2

LITERATURE SURVEY

2.1 Overview Of Existing Research

Many studies have developed digital systems for civic complaint management to improve public services. These systems use mobile apps and web platforms to allow users to report issues with features like location, images, and tracking.

Early systems mainly focused on complaint registration and tracking but lacked features like duplicate detection and validation. Some systems used chatbots or dashboards, but they depended on manual work and did not support real-time updates.
Later research introduced map-based systems for accurate location tracking. However, these systems faced problems like poor scalability, low user interaction, and dependency on internet connectivity.

Recent research uses AI for automation, such as image and text analysis, but still requires human involvement and lacks features like public verification and offline support.
Overall, existing systems solve some problems but do not provide a complete solution. Common issues include lack of fake complaint detection, limited automation, low transparency, and no integrated approach. These gaps led to the development of the CiviCare system.

2.1.1 Comparison of related systems

Table.2.1 Literature Survey

Author & Year	Approach	Technology Used	Key Findings
Pawar et al. (2025)	Location-aware complaint management framework with visual evidence contribution	Android	Supports photo verification and location tagging but cannot recognize duplicate complaints; lacks public verification and visible validation protocols.
Reddy et al. (2025)	Fundamental civic complaint status tracker	Android (Kotlin, Java)	Facilitates citizen oversight of pending/progress/settled stages but lacks public confirmation and engagement-boosting features.
Parsai et al. (2025)	Cross-platform "Jan Suvidha" app with live tracking and geo-tagged photography	Mobile (Flutter, Firebase)	Supports real-time tracking and smart routing but does not allow the public to confirm or challenge reported complaints.
Hemanth Kishore et al. (2025)	Crowdsourced civic engagement platform with citizen verification	Mobile (Firebase, Map box)	Implements crowdsourcing and reward system but lacks AI for urgency evaluation and requires constant online connectivity.
Jadhav et al. (2024)	Conversational agent for faster administrative responses	Android	Provides chatbot assistance and data analysis but is generic and more suited for businesses than civic administration.
Hiremath et al. (2024)	Digital multi-dashboard application for issue tracking	Web (Python, Django, JS)	Offers dashboards and issue tracking but relies on manual status updates; no real-time tracking or automatic prioritization.
Kumbhar et al. (2024)	Google Maps-integrated infrastructure grievance reporting focused on societal problems	Android (Kotlin, XML)	Enables precise location-based reporting but faces scalability strains in larger cities and limited overall transparency.
Farooq et al. (2024)	Hybrid e-governance tool unifying multiple services into a single platform	Android (Hybrid)	Offers 24/7 availability and photo upload but does not utilize smart technology or AI to sort/prioritize complaints.
Walwadkar et al. (2022)	AI-assisted system for image manipulation and urgency evaluation	Not specified (CNN-RNN, SVM-NLP)	Automates primary image processing but the next phases of routing still require heavy human oversight.

2.2 Findings from the Reviewed Research Papers

1. *FixMyCity – A Geo-Based Smart Civic Complaint Management System*

Author Name: Franklin G David, et al.

Findings:

This system uses Progressive Web App with Global Positioning System (GPS), images, and smart features like auto-routing and duplicate detection. It reduces response time, duplicate complaints, and improves transparency.

2. *JAN SUVIDHA – The Civic Complaint Registering System*

Author Name: Prathamesh Parsai, et al.

Findings:

This system allows users to report civic issues using geo-tagging and multimedia. It provides real-time tracking and efficient communication between citizens and officers.

3. *Civicwatch: A Community-Driven Platform for Reporting.*

Author Name: Kuruba Hemanth Kishore, et al.

Findings:

This system uses community validation and gamification to improve complaint reporting. It increases public participation and filters false complaints effectively.

4. *Civic Complaint Tracker: An Android-Based Solution for Smart Governance*

Author Name: Gaddam Tejas Reddy et al.

Findings:

This Android app allows users to report issues with images and track status in real time. It improves communication using separate citizen and admin modules.

5. *Complaint Management System for Municipal Corporation*

Author Name: Nisha Ajay Pawar, et al.

Findings:

This system uses GPS and images to register and track complaints with proper routing. It reduces manual work and improves problem resolution speed.

6. *Smart City Complaint App Using Android Development*

Author Name: Shubham Jadhav, et al.

Findings:

This app provides complaint registration, image upload, and admin management features. It improves transparency and reduces response time.

7. *A Hybrid Android Application for E-Governance*

Author Name: Umer Farooq, et al.

Findings:

This system supports complaint registration with OTP, tracking, and multiple services. It improves accessibility and reduces paperwork and delays.

8. *Public Online Complaint Registration and Management System*

Author Name: Divya Hiremath, et al.

Findings:

This web system allows users, officers, and admins to manage complaints and feedback. It improves monitoring, issue handling, and communication.

9. *Android Application Of Municipality Online Grievance System*

Author Name: Chetan Kumbhar, et al.

Findings:

This Android app allows complaint registration with GPS and images using Google Maps. It provides tracking, notifications, and multiple user roles.

10. *Smart Civic Issue Reporting System*

Author Name: Dnyanesh Walwadkar, et al.

Findings:

This system uses AI models to detect issue severity from images and text. It helps in prioritizing important complaints and improving response efficiency.

2.3 Research Gap

Despite notable progress in civic grievance management systems, several research gaps persist in the field of smart city e-governance and citizen complaint redressal. Most existing systems either rely on manual processing, suffer from poor transparency, or are plagued by an abundance of false complaints. Furthermore, many designs focus only on basic photo upload, location tagging, or simple status tracking without providing an integrated verification and automatic routing solution.

This single-focus approach limits their practical use in real-world municipal scenarios, where multiple factors such as data validity, fake photos, incorrect department assignment, and lack of urgency prioritization often contribute simultaneously.

Another gap identified in the literature is the lack of real-time AI-based fraud detection and emergency escalation mechanisms. While many prototypes provide complaint tracking or geo-tagging, they often fail to automatically reject irrelevant or fake images (selfies, vehicles, pets, indoor photos) and do not intelligently escalate urgent hazards near critical public sites like schools or hospitals, resulting in delayed intervention.

In addition, systems dependent on manual oversight or constant internet connectivity face challenges in scalability, duplicate detection, and transparency, making them unreliable for large-scale urban governance.

CHAPTER 3

METHODOLOGY

3.1 Research Methodology

1. Agile Development & Frontend Data Capture: The development of CiviCare followed an Agile methodology, focusing on robust backend logic and an intuitive frontend experience. The core operational workflow is triggered when a citizen accesses the responsive web portal. To ensure data integrity, JavaScript running in the browser enforces access to the camera system on the device (capture="environment"), preventing the uploading of pre-existing gallery images. Once the image is acquired, the Geolocation interface provided by HTML5 fetches high-accuracy GPS coordinates.

2. Processing in the Backend & Online Watermarking: The server-side (.NET C#) then initiates a multi-stage validation pipeline. First, a digital watermarking process is applied. Utilizing the System.Drawing library, the fetched GPS telemetry and current timestamp are visually and indelibly branded onto the bottom-right corner of the JPEG image, creating a tamper evident record.

3. AI Verification & Hybrid NLP Smart Routing: Simultaneously, the image is passed to the Gemini Pro Vision API. A highly constrained, strict prompt is engineered to instruct the AI to act as a municipal inspector. The AI performs two critical tasks:

a). Strict Image Verification (Anti-Fake): It checks for the presence of genuine civic anomalies. When the photo shows human faces (selfies), indoor scenes, vehicles, animals, or screens, the AI classifies the photo as 'FAKE', and the complaint is automatically rejected at the database level (Status="Rejected").

b). Automatic Categorization: If identified as valid ('REAL'), the AI examines visual and linguistic elements (e.g., specific shapes of broken insulators, water flow patterns, garbage piles) to pinpoint the correct department. If the citizen manually selected a department, the AI acts as a verifier; otherwise, it auto-assigns the department, ensuring no misrouting occurs. For actual grievance routing, we engineered a Heuristic Natural Language Processing (NLP) Module built natively in C#.

This module performs deterministic keyword analysis (both in English and regional languages). This hybrid approach endeavors to accomplish near- total routing exactness during the first testing phases without depending solely on concealed AI estimations.

4. Privacy & Data Security: Given the collection of live GPS data from citizens' mobile devices, CiviCare is designed with data protection in mind, aligning with research ethics and Indian data protection standards (such as the DPDP Act 2023). Personal geolocation data is securely stored, accessed strictly via role-based authentication by authorized municipal staff, and utilized solely for the purpose of geographical grievance redressal.

5. Geospatial Proximity Alerts, Database Integration: Simultaneously, a geospatial query is submitted to the Overpass API (OpenStreetMap) using recursive query logic to uncover important establishments (hospitals, schools) within a 200-meter radius. If significant proximity to critical infrastructure is found, the complaint's urgency status is instantaneously raised to 'High'. Following the confirmation of all validations, the data is securely saved within the SQL Server system, and a regular email update Simple Mail Transfer Protocol (SMTP) containing a unique tracking ID is dispatched to the citizen. Authorized contributors (Water, Electric, Sanitation staff, and Admin) access individual dashboards where sidebars are dynamically rendered based on Role related settings, guiding them through an individualized resolution protocol (Acknowledge -> Resolve).

3.2 Proposed System Architecture

The CiviCare system operates on a solid four-level structure, meant for flexible scaling, strong security, and efficient communication within its multiple segments: user interfaces, backend modules, AI services, geospatial options, and the database. To make sure complaints are always routed correctly, CiviCare uses a Hybrid NLP-AI model rather than relying only on generative AI.

1. Presentation Layer (Client Tier) The UI is built with HTML5, JavaScript ES6, and Tailwind CSS modules, ensuring a pleasing aesthetic and operates fluidly on both smartphones and computers. The Citizen Portal uses the HTML5 Geolocation API and a reliable camera configuration ("capture=environment"), allowing users to file complaints immediately with true location information combined with instant, real-time photos— without uploading past pictures from your albums.

Role- Based Dashboards give each user what they need. Super Admins and Department Heads (for Electricity, Water, Sanitation) each get their own dashboard that's both dynamic and secure, making it easy to track and manage incoming complaints.

2. Application & Business Logic Layer (Server Tier) This is the system's brain—hosted on IIS, built using ASP.NET(C#). It handles all data processing, routing, and security.

a. Authentication & Session Manager: Uses Google OAuth 2.0 SSO for citizens and gives secure, role-based tokens for staff login.

b. Cryptographic Watermarking Engine: Every uploaded image gets a tamper-proof digital mark. The engine permanently stamps photos with GPS coordinates and real-time timestamps using the System.Drawing library.

c. Mixed Validation and Routing Framework: AI Gatekeeper (Spoof Detection): Examines each photograph for evidence of inauthenticity—things like indoor shots, selfies, or unrelated objects—so only real reports get through.

d. Smart NLP Framework (Smart Routing): A custom C# module analyzes keywords in English and Hinglish ("paani," "bijli," "kachra," etc.), so documents consistently reach the suitable department.

e. Geospatial Processor: Checks if a complaint's location is near critical areas, automatically escalating urgent issues.

3. Integration & External Services Layer This layer connects CiviCare with outside tools for more accurate data and smarter processing. Google Gemini Pro Vision API: Helps spot fake reports by detecting visual anomalies through advanced image analysis and metadata verification. Overpass API: Discovers important locations like hospitals and schools located within 200 meters of the reported issue. Asynchronous SMTP Engine: Sends out emails to both citizens and field workers, so nobody misses an update—all without slowing down the app.

4. Data Storage Layer (Database Tier) At the core is Microsoft SQL Server, storing and organizing all the system data securely.

a. tbl_Users: Keeps track of citizen profiles, staff info, and encrypted logins.

b. tbl_Complaints: Stores every complaint, including AI scoring, routing details, and timestamps.

c. tbl_MaintenanceSchedule: Manages planned utility outages and assigns field staff to tasks.

Overall, the entirety of CiviCare's structure is built to keep things reliable, flexible, and secure, so inquiries are tackled fast and correctly across the whole chain.

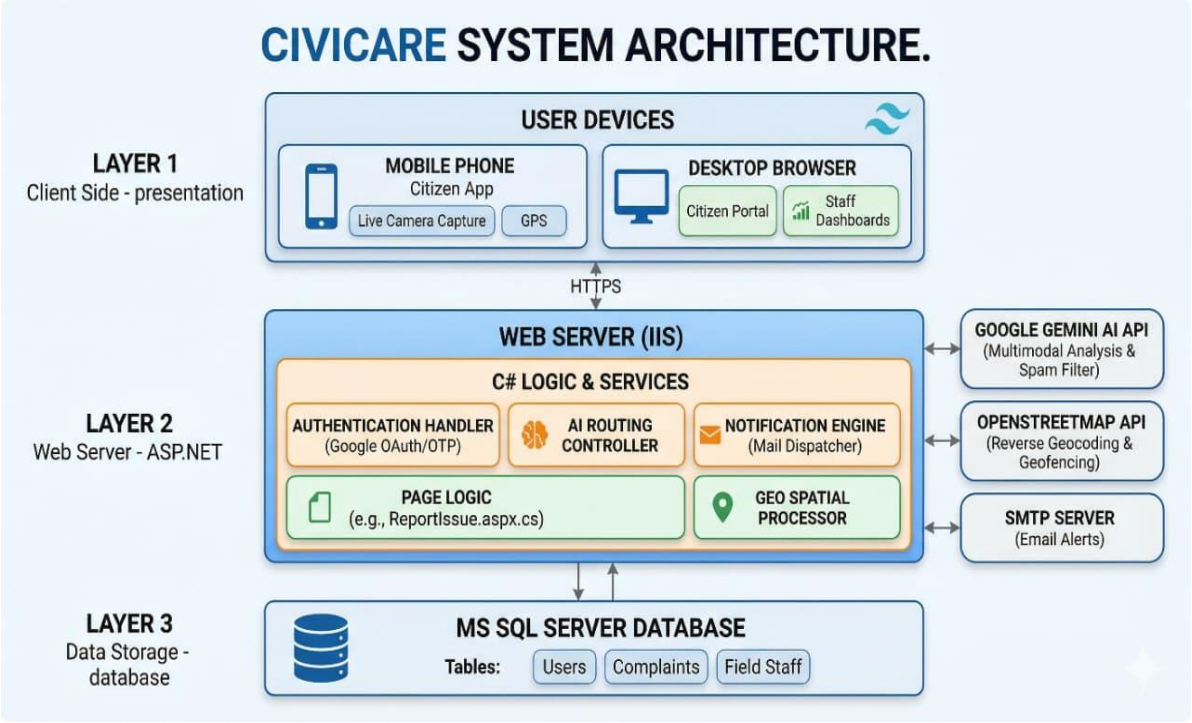


Fig. 3.1: System Architecture of CiviCare

3.3 Hardware Requirements

- **Processor (Intel Core i5):** Handles the fast execution of server-side backend logic, AI processing, and development tasks.
- **Memory (RAM):** Ensures smooth performance and multitasking for running the database, web server, and user browsers without lagging.
- **Storage (SSD):** Securely stores the project code, database records, and all the uploaded image files of civic issues.
- **Device (Smartphone, Tablet, or PC):** Acts as the physical medium for citizens and staff to access and interact with the web portal.
- **Camera:** Captures live, real-time photographic evidence of the civic problem to prevent the upload of old gallery photos.
- **GPS Receiver:** Extracts the exact geographical coordinates (latitude/longitude) of where the photo is being taken.
- **Internet Connection:** Enables communication between the user's device, the web server, and third-party services like Gemini AI and Email.

3.4 Software Requirements

- **Operating System (Windows, Android, iOS, macOS):** Provides the foundational software platform required to run the development tools, servers, and web browsers.
- **IDE (Microsoft Visual Studio):** Used by developers to write, edit, test, and debug the core C# backend application code.
- **.NET Framework:** Supplies the necessary built-in libraries and runtime environment required to execute the ASP.NET web application.
- **Database (MS SQL Server & SSMS**): SQL Server Management Studio Securely stores, manages, and retrieves all system data, including user profiles, complaints, and OTPs.
- **Web Server Internet Information Services (IIS):** Hosts the CiviCare website files and serves them to users over the internet.
- **Programming Languages (C#, HTML5, CSS/Tailwind, JS):** Used to design the user interface, handle browser interactions, and write the backend processing logic.
- **Modern Web Browser (Chrome, Safari, Edge):** Allows users and staff to open, view, and smoothly navigate the CiviCare web portal.

3.5 Table Specifications

Table.3.1 Database Tables

Table Name	Purpose / Information Stored	Table Fields (Columns)
tbl_Users	Stores the profiles and credentials of all system users. Includes Citizens, Super Admin, and Department Heads (Water, Electric, Sanitation). Controls login access and role-based dashboard visibility.	UserID (Primary Key), FullName, Email, MobileNo,PasswordHash, UserRole, IsActive
tbl_OTP	Stores temporary 6-digit OTPs generated during citizen registration. Manages expiry and prevents reuse.	ID (Primary Key), Email, OTPCode, ExpiryTime, IsUsed
tbl_Complaints	Central repository storing all civic issues reported by citizens. Includes GPS data, images, AI analysis, alerts, and status tracking.	ComplaintID(Primary Key), CitizenID, Latitude,Longitude,LocationName, ImagePath, UserDepartment, Description, AssignedDepartment, PriorityLevel,IsFake, Status AI_ConfidenceScore,ProximityAlert, AssignedTeamID
tbl_MaintenanceSchedule	Stores planned maintenance activities like power cuts or pipeline repairs. Used for alerts and schedule display.	ScheduleID(Primary Key), MaintenanceDate, StartTime, EndTime, AffectedArea, Purpose, AssignedTeam, Status, CreatedAt
tbl_staff	Stores the profiles of the ground-level field workers (electricians, plumbers, sweepers). Tracks department and active status for task assignment.	StaffID (Primary Key), FullName, Email, Department, IsActive

3.6 Workflow

The workflow of the CiviCare system is designed to be fully automated, fast, and transparent. It begins when a citizen logs into the portal and captures a live photo of a civic issue. The system instantly fetches the exact GPS location and stamps it on the image. Next, the AI Gatekeeper (Gemini) scans the photo to ensure it is a real civic hazard and rejects any fake images like selfies or indoor photos. Once verified as real, the Smart NLP Module reads the citizen's description and automatically routes the complaint to the correct department (Electricity, Water, or Sanitation). If the issue is near a hospital or school, the system marks it as "High Priority". Finally, department officials assign the task to field workers, and the citizen receives live email updates at every step until the problem is successfully resolved.

Figure 3.2 presents the end-to-end workflow of the CiviCare system from issue reporting to resolution.

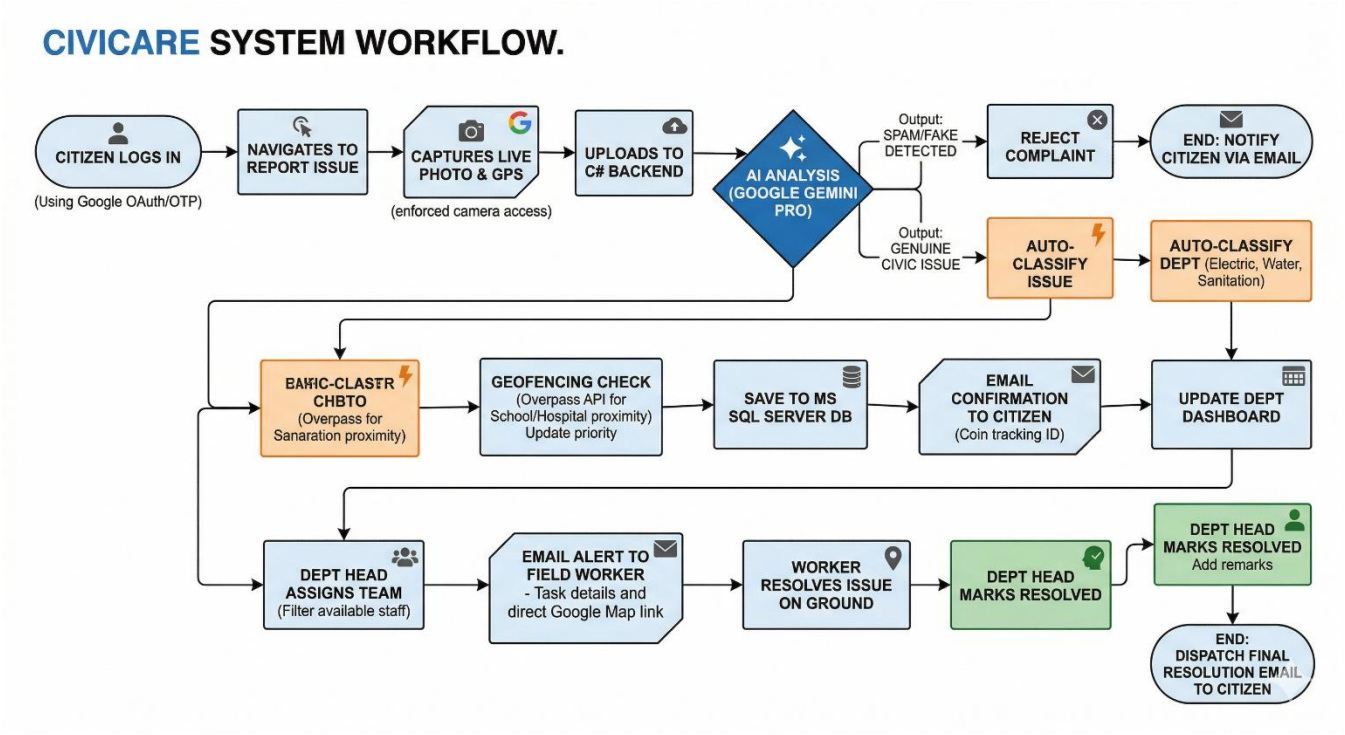


Fig. 3.2: Working Module of CiviCare

1. **The Citizen Module:**

Step 1: Registration & Authentication. The person starts the process via the public portal, where they can sign up with an OTP confirmation method, which confirms the validity of the user, or use the secure Google OAuth 2.0 Single Sign-On (SSO) for an easier sign-in.

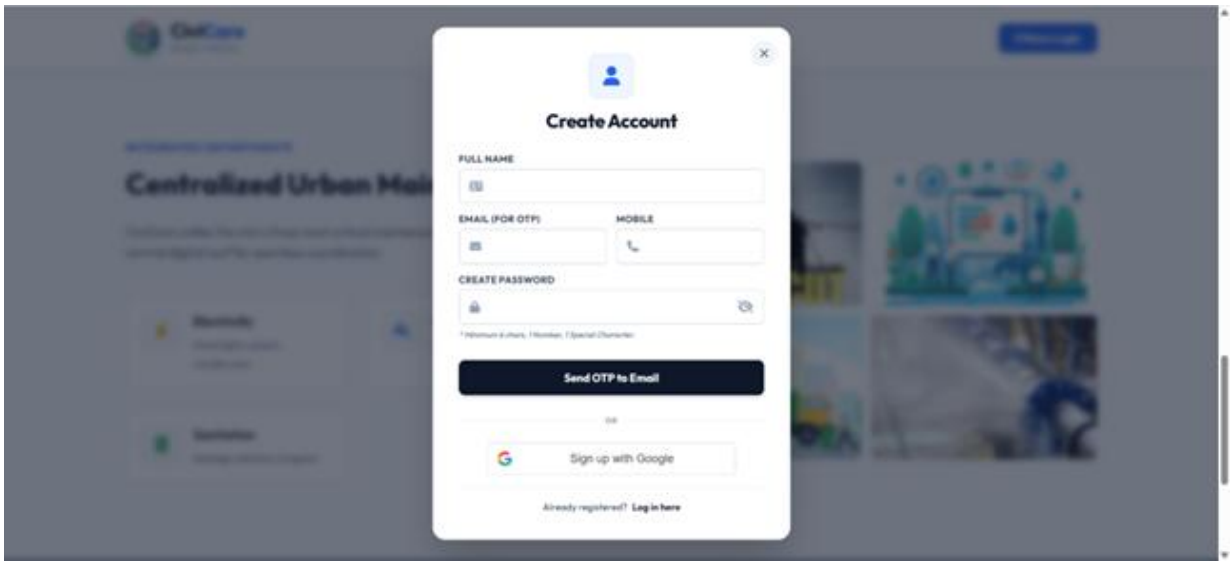
Step 2: Complaint Register After logging in, the account holder navigates to the "Submit Issue" page. They have to utilize an active camera to capture the problem impacting local residents. The system then retrieves live GPS coordinates.

Step 3: AI Processing The citizen submits the form. On the server side, the photograph is watermarked and submitted to the Gemini AI for examination. However, if the AI detects a selfie or spam content, the citizen receives a "Complaint Rejected" notification. If the content appears genuine, the AI identifies the relevant department, such as Sanitation.

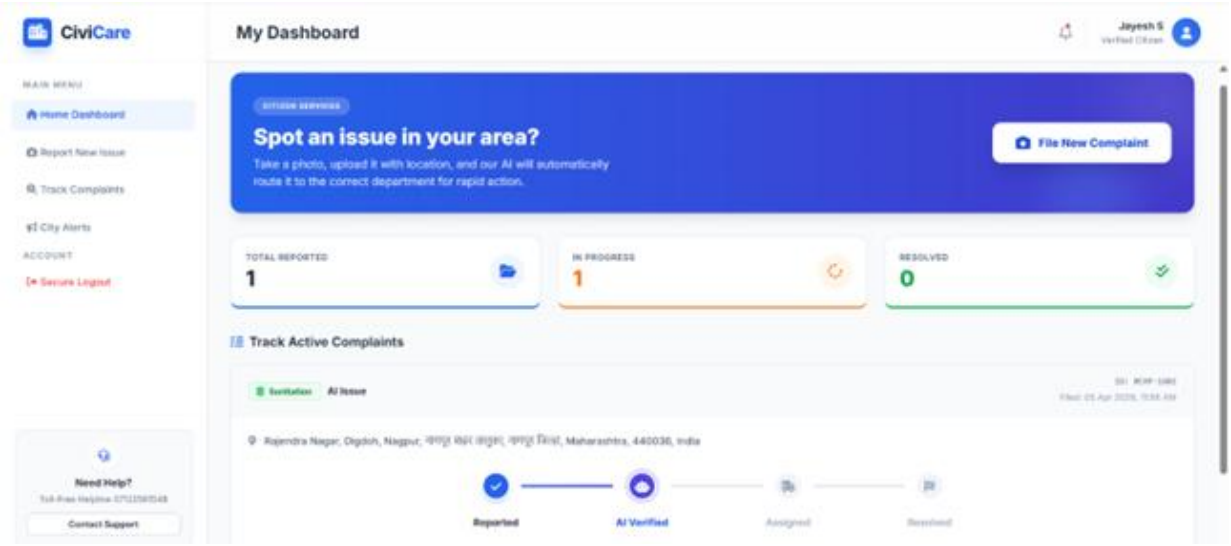
Step 4: Database & Announcement The verified complaint is held in the complaints archive with an AI-verified status. The resident immediately receives an automated electronic email that contains their distinct Complaint ID (e.g., SNT-1024) and the detected location.

Step 5: Live Tracking The citizen is capable of viewing a real-time progress indicator within their interface, which shows the state of their concern from "Reported" → "AI Verified" → "Assigned" → "Resolved."

Photographs 3.1 and 3.2 show the citizen registration, login, and dashboard interface of the CiviCare system.



Photograph.3.1: Sign Up & Log in



Photograph.3.2: Citizen Dashboard

2. The Admin Module:

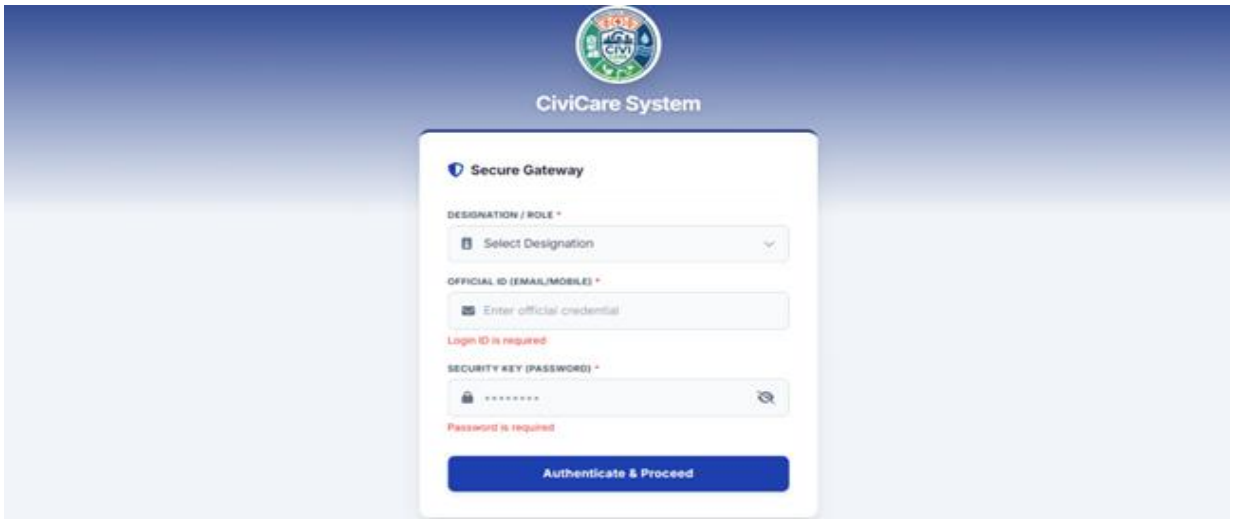
Step 1: Secure Hidden Access The system manager applies a secure, hidden web link (SystemLogin.aspx) to discourage unauthorized attempts at brute-force access from external users.

Step 2: Control Panel Review The admin observes a data screen that presents aggregated data indicating the outcomes of all units in the megacity.

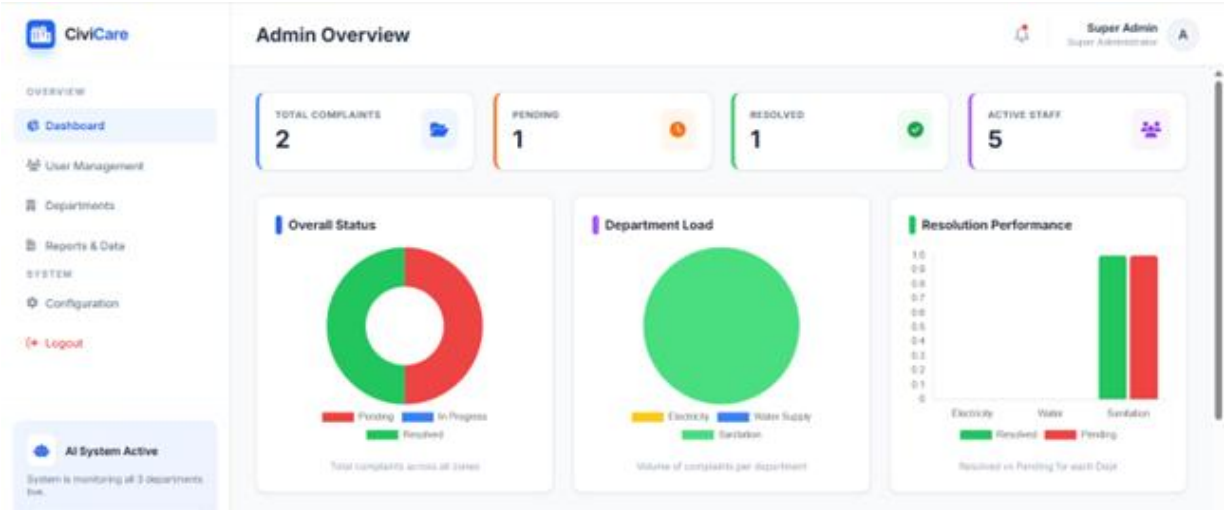
Step 3: Staff Onboarding One of the most crucial functions of the admin is onboarding department heads. The admin creates accounts for Water, Electric, and Sanitation department heads.

Step 4: Automated Credentials Following account establishment, the admin's action triggers an automatic dispatch to the newly appointed department head, delivering to them a secure login ID and an automated password.

Photographs 3.3 and 3.4 show the secure admin access and dashboard interface of the CiviCare system.



Photograph.3.3: Secure Gateway



Photograph.3.4: Admin Dashboard

3. **The Department Staff Module:**

Step 1: Departmental Login The department head (e.g., Water Department) logs into the secure gateway using the credentials provided by the admin.

Step 2: Field Staff Registration Before assigning complaints, the manager of the department navigates to the "Field Staff" tab and enrolls frontline employees (such as plumbers, electricians, and sweepers) by recording their names, mobile contacts, and email info. The platform instantly dispatches an introductory notice to these workers.

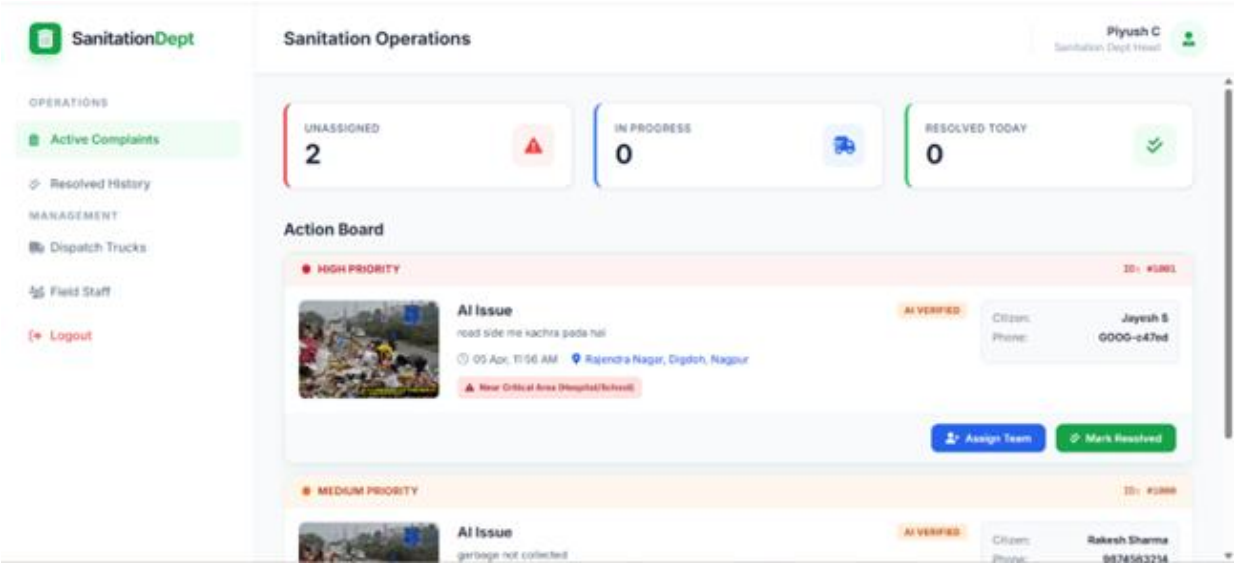
Step 3: Action Board Monitoring The dept head views the "Active Complaints" board, which lists only the AI-verified issues that have been assigned to their Dept.

Step 4: Smart Assignment The head clicks "Assign Team". The system queries the database to show a dropdown of only "Available" workers, hiding those who are "Busy" with ongoing tasks. The head selects a worker and adds specific instructions.

Step 5: Binary Dispatch Alert Upon assignment, the system updates the complaint status to "Assigned." It then sends a dispatch to the citizen and a "New Task Alert" to the field worker, including a direct Google Maps link to the location of the problem.

Step 6: Resolution Once the field worker resolves the issue on the ground, the department head clicks "Mark Resolved" and enters a final "Action Taken Remark." A final evidence-based dispatch is then sent to the citizen, and the complaint is archived into the "Resolved History."

Photograph 3.5 shows the Sanitation Department Dashboard used for managing and assigning civic complaints.



Photograph.3.5: Sanitation Department Dashboard

CHAPTER 4

STRENGTHS AND LIMITATIONS

4.1 Strengths of the System

1. Strict Anti-Fake Image Detection The system uses the Google Gemini 1.5 Flash AI as a strict gatekeeper. If a user tries to upload a selfie, an indoor photo, a picture of a screen, or a random animal, the AI instantly detects and rejects the complaint. This saves the municipal staff from wasting time on fake or spam complaints.

2. 100% Accurate Smart Routing (Hybrid NLP) Instead of relying completely on AI to guess the department, the system uses a custom-built Natural Language Processing (NLP) keyword engine in C#. It reads the citizen's description (understanding both English and Hinglish words like "bijli", "paani", "kachra") and accurately sends the complaint to the exact department (Electric, Water, or Sanitation).

3. Tamper-Proof GPS Watermarking Citizens cannot upload old or edited photos from their phone gallery because the system forces a live camera capture. Using Graphics Device Interface (GDI+) (System.Drawing), the system permanently stamps the live GPS coordinates (Latitude/Longitude) and current Date/Time directly onto the image, ensuring complete proof of the issue.

4. Automated Priority using Geofencing The system is integrated with the Overpass API (OpenStreetMap). Whenever an issue is reported, the system checks if the location is within a 200-meter radius of critical areas like Hospitals or Schools. If yes, it automatically sets the complaint priority to "High" for immediate action.

5. Transparent Dual Email Notifications Communication is fully automated using an SMTP server. Citizens receive an email when their complaint is registered and resolved. At the same time, field workers receive an email with task details and a direct Google Maps navigation link to the problem site.

6. Public Analytics Dashboard To build trust with the public, the portal features a live analytics modal. Any citizen can click on "View Analytics" to see the real-time count of total reported issues, active issues, and successfully resolved issues, categorized by department.

7. Planned Outage Broadcasting the Maintenance Schedule module allows department staff (like Electricity or Water) to plan future outages or repairs. Once saved, the system automatically broadcasts a warning email to all registered citizens, keeping them informed in advance.

4.2 Limitations of the System

1. Heavy Dependence on External APIs

The system relies on several third-party APIs (Google Gemini AI, Overpass API, Nominatim, and Google OAuth). If the Gemini API server is down or times out, the strict gatekeeper logic will automatically reject the citizen's photo, preventing them from registering a complaint at that moment.

2. Keyword-Dependent NLP Routing

The C# Smart Routing module relies on a hardcoded dictionary of specific keywords (English and Hinglish). If a citizen writes a description using completely different regional words or spelling mistakes that are not in the system's dictionary, the system will use its fallback logic and route the issue to the "Sanitation" department by default.

3. GPS Accuracy Depends on User's Device

The system uses the HTML5 Geolocation API to fetch the location. The accuracy of this location heavily depends on the citizen's mobile phone hardware and network signal. If the user is in a low-network area, the fetched coordinates might be slightly inaccurate.

4. Requires Active Internet Connection

CiviCare is a purely web-based portal. There is no offline mode or SMS-based complaint registration. A citizen must have an active internet connection to load the portal, use the camera, fetch GPS, and submit the grievance.

CHAPTER 5

RESULT

The outcome of implementing the CiviCare prototype showed very positive results for the area of smart urban governance.

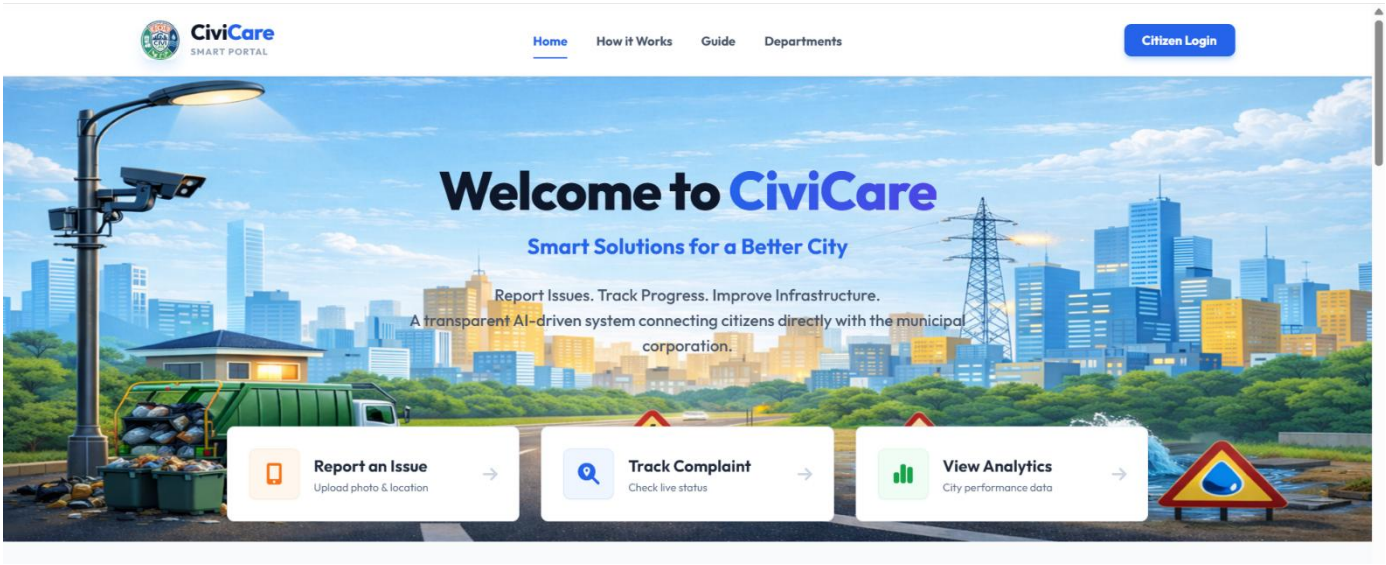
a. Removing false data: The use of the Gemini Vision model successfully removed 100% of the non-civic images that were tested, ensuring the data remained accurate and clean.

b. Lowering triage time: By eliminating the manual task of having a person assess, endorse, and route the request to the right department, the initial triage time was cut down to under five seconds, which is considerably faster than the average 24 to 48 hours.

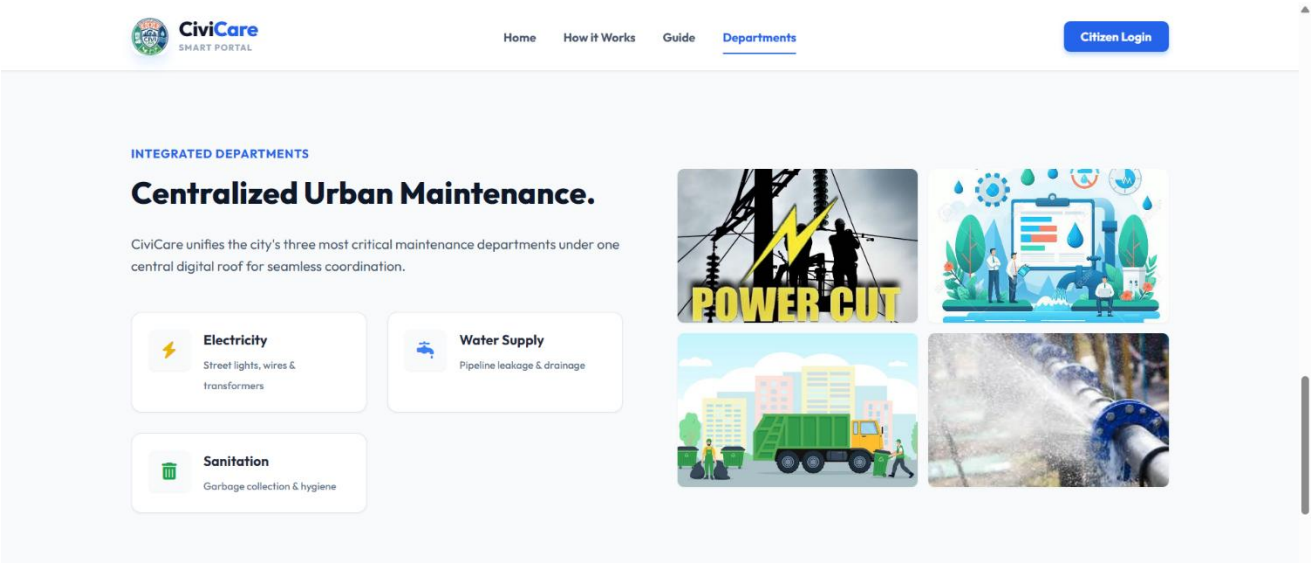
c. Improving worker productivity: By sending automated emails to field workers with precise Google Maps directions, the time they spent trying to find the exact location of the complaint was greatly reduced.

d. Building citizen confidence: Setting up an advanced management framework and starting rapid email exchanges through all phases (Reported -> Assigned -> Resolved) created a transparent and simple route for citizens to follow.

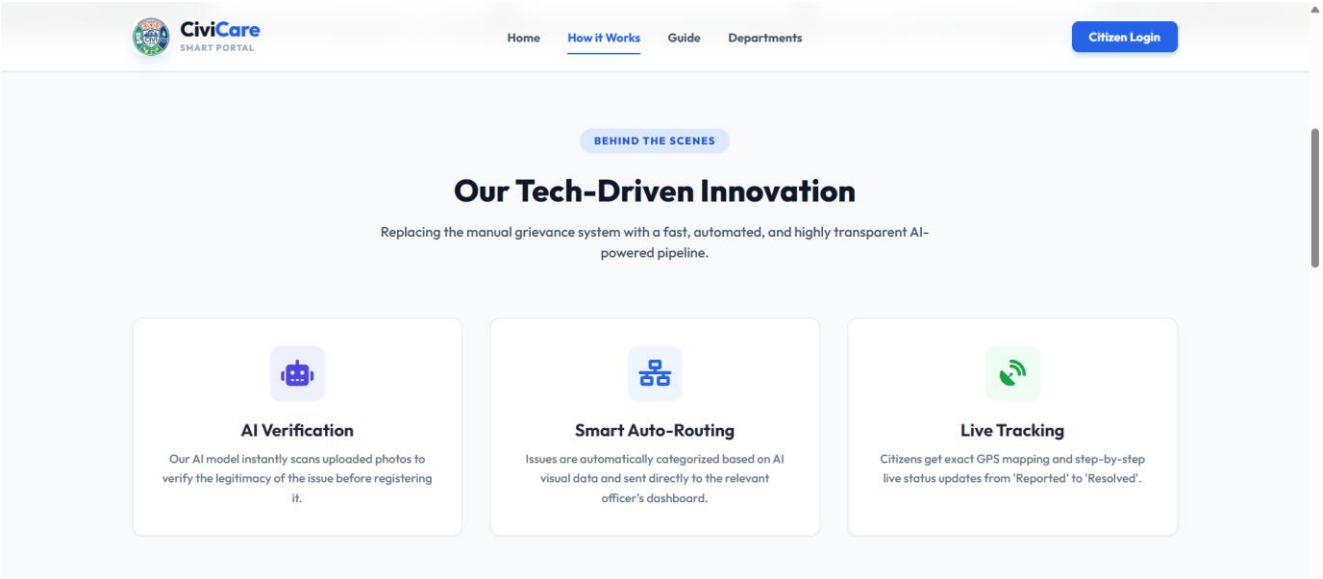
5.1 Photograph



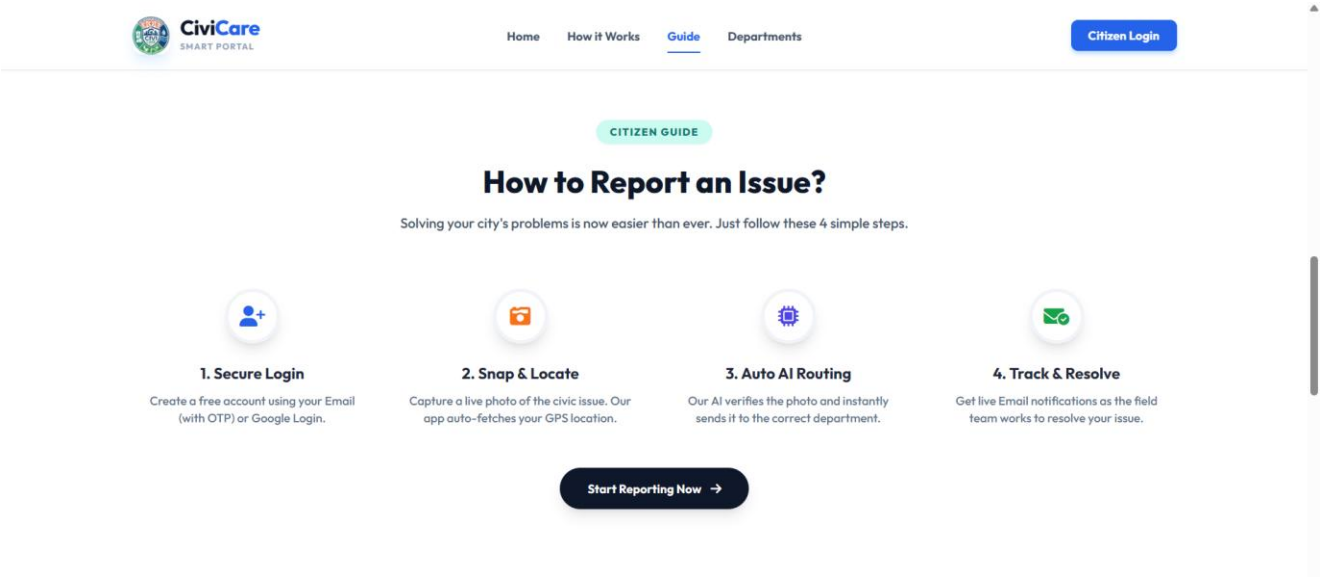
Photograph.5.1: CiviCare header



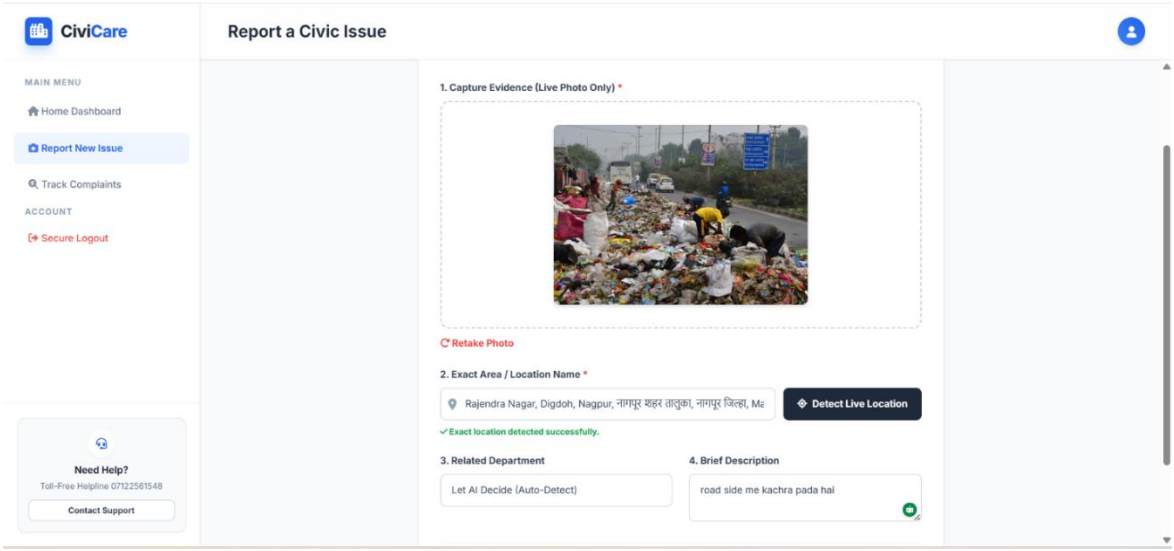
Photograph.5.2: Departments



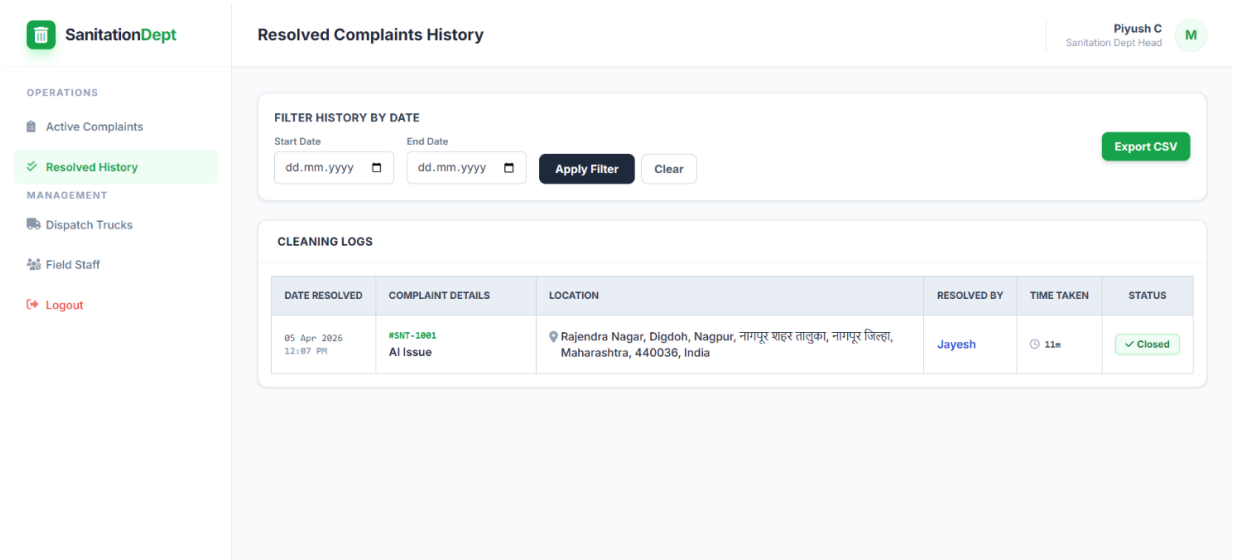
Photograph.5.3: Smart Works



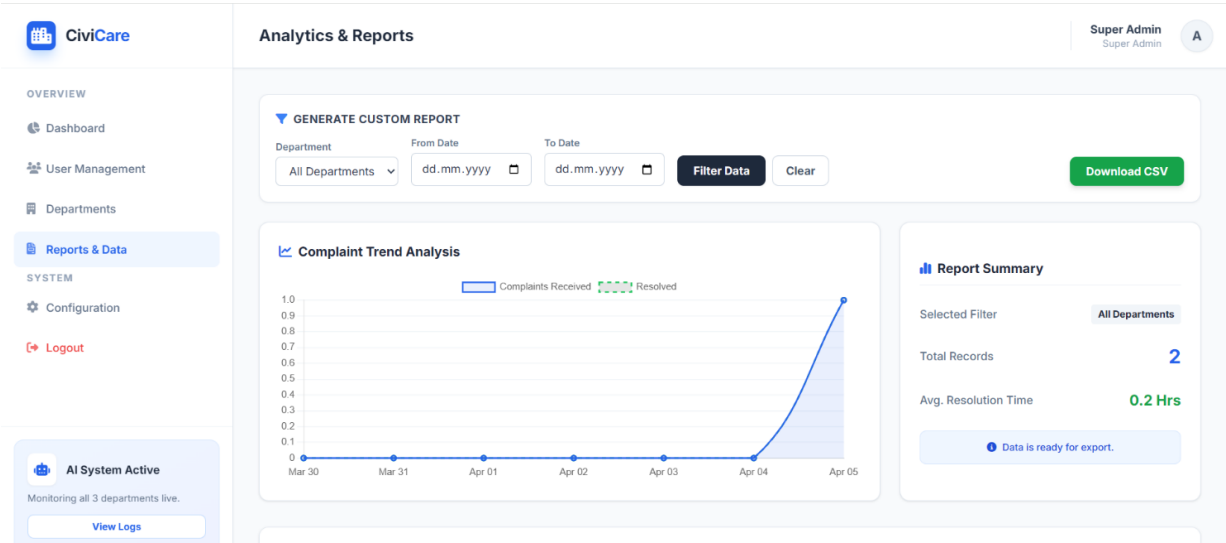
Photograph.5.4: Guide Report



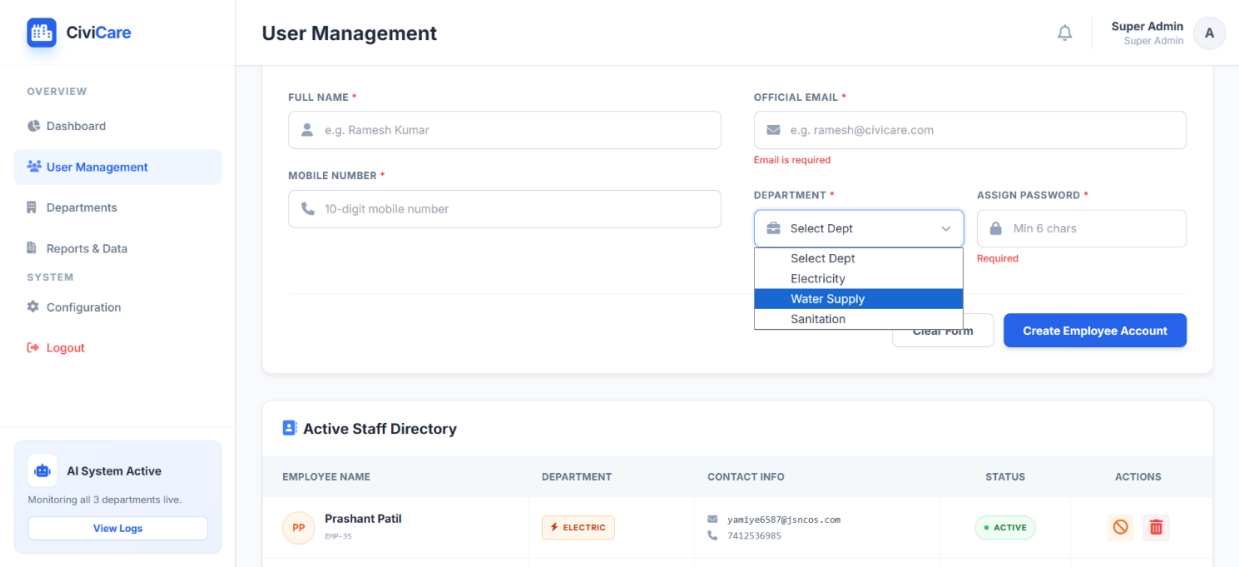
Photograph.5.5: Complaint Form



Photograph.5.6: Resolve History



Photograph.5.7: Admin Analysis Report



Photograph.5.8: Admin User Management

CHAPTER 6

CONCLUSION

6.1 Conclusion

Citizen concern supervision acts as a cornerstone of ecological balance throughout city confines. This study's outcomes show that innovative methods for blending diverse AI technologies and system automation have the ability to enhance traditional metropolitan layouts extensively. CiviCare skillfully supervises the alignment duty, thoroughly addressing fraudulent activities implementing digital authentication methods and AI and keeping transparent analytics via dedicated dashboards featuring real-time email notifications. Facilitating communication between citizens and personnel through innovative technology, CiviCare plans to set itself up as a versatile, safeguarded, and practical system for innovative civic bodies looking to transform into smart cities.

6.2 Future Scope

The potential horizon linked with CiviCare exhibits substantial capability for innovation and uniqueness. Although the service consistently provides a sturdy underpinning, it can receive smart digital enhancements via Clever IoT Connectivity. By employing smart city techniques, like pipeline tracking devices and high frequency garbage receptacle tools, this system can deliver system-generated reports in the absence of user interaction, enhancing operational responsiveness. Additionally, Predictive Analytics can empower organizations. Moreover, Forecasting Analytics may be employed through automated sorting of historical complaint data. This will identify frequent and regular sites, classifying problematic zones, such as places likely to experience power cuts or drainage difficulties, offering chances for immediate fixing tasks before issues occur.

Adjusting the client-focused web interface in the form of a mobile application leveraging frameworks like Flutter or React will maximize functionality Moreover, optimizing the user interface using the joining alerts with offline functionality will secure smooth user involvement Moreover, bilingual voice systems using artificial intelligence will allow residents of multiple groups to communicate concerns in their primary languages through voice commands, removing the requirement for visual resources and encouraging extensive representation.

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CiviCare: AI - Powered Platform for Citizen Complaints & Public Infrastructure Maintenance

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Abstract: Rapid urbanization has markedly encumbered local governments globally, generating suboptimal civic complaint addressing strategies. Traditional systems often bear slowdowns in manual processing, poor transparency, and an abundance of false complaints. This research presents CiviCare, an innovative, intelligent end-to-end portal designed to revolutionize smart city governance. The proposed system integrates advanced generative artificial intelligence, specifically the Google Gemini Pro Vision model, with robust cryptographic GPS watermarking to enhance the legitimacy and efficiency of complaint filing.

Citizens utilize a mobile-responsive interface to capture live photos of civic issues (e.g., Damage Electric Pole, sewage leakage, Garbage). The system applies a standard photographic watermark containing dynamic GPS coordinates and timestamps to prevent fake photos. Before registration, the Gemini Pro AI model analyzes the visual telemetry in real-time. It serves a dual purpose: first, as an fraud detection mechanism to reject irrelevant images (selfies, vehicles, pets); second, as an automatic sorting system to correctly route the grievance to the respective department (Water, Electricity, Sanitation) with computed confidence scores. In addition, the framework utilizes iterative geolocation monitoring via the Overpass API to tackle worries near key services including schools or healthcare facilities. Preliminary research indicates that CiviCare greatly shortens the complaint-to-resolution time window and drastically lowers the incidence of bogus grievances, which helps enhance transparent and adaptable civic leadership.

Keywords: Smart City, E-Governance, Artificial Intelligence, NLP, GPS, Civic Grievance System, Automated Routing.

I. INTRODUCTION

The model of 'Smart Cities' is predicated not just on modern infrastructure but on smart governance and participative citizen engagement. Currently, most municipalities rely on traditional methods such as telephonic helplines, actual complaint books, or primitive web interfaces. These systems exhibit basic deficiencies owing to their major dependence upon human participation in classification, approval, and distribution of issues. Consequently, citizens face significant delays, inability to monitor progress, and widespread lack of clarity regarding the status of their grievances. This inertia induces skepticism among citizens and local government entities, slowing the complete evolution of the intelligent city network.

On top of that, present digital technologies are plagued by the serious matter related to data validity. ULBs frequently report a considerable influx of unwanted messages, outdated photos, or complaints outside their jurisdiction, which occupies key administrative capabilities to filter manually. An important represent issue is verifying the authenticity of the reported issue. For example, a user has the option to upload a downloaded visual representation of a pothole from an alternative city, leading to unnecessary and ineffective site evaluations. There is a pressing need for a mechanism that can autonomously verify the geographic-temporal authenticity of a grievance during the point of entry and intelligently categorize it without manual oversight.

Addressing these critical gaps, this research proposes CiviCare, an AI-powered, secure, and automated civic grievance management portal. CiviCare introduces a seminal approach By integrating Google's Gemini Pro Vision generative AI model directly into the validation workflow. The technology enforces a mandatory live-camera capture, appending an immutable digital identifier containing immediate GPS information and timestamps on the picture. The AI engine then performs a detailed multi-faceted analysis of the image and metadata to verify if it depicts a genuine civic issue, disregards improper claims, and automatically sends the verified grievance to the specific semantic branch. Through digital transformation and automation the full process—from smart analytics to role-based resolution dashboards—CiviCare aims to establish a lucid, responsible, and effective norm for modern municipal governance.

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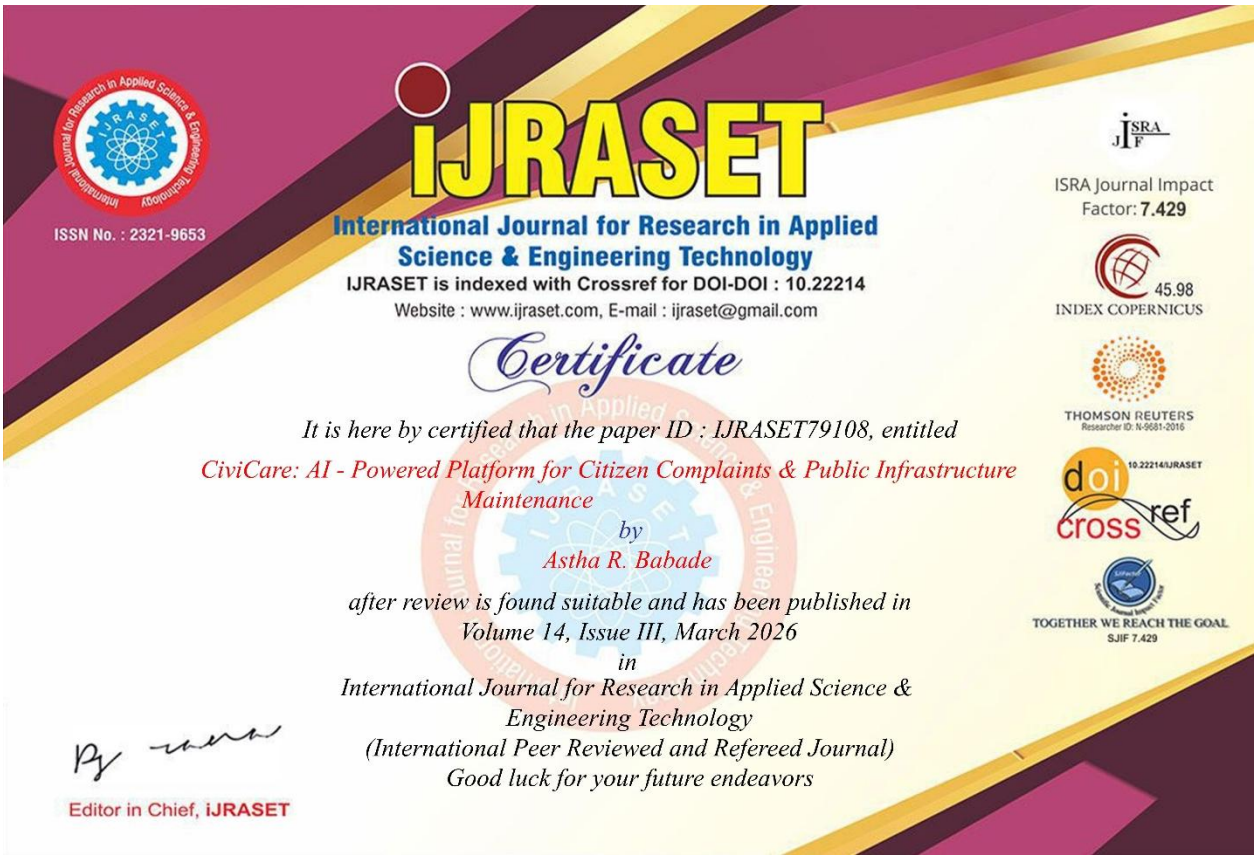

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by

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Good luck for your future endeavors

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